Supplementary Material

INS-Aided Deep Tracking of GPS and NavIC Signals Under Simulated Aircraft Maneuvers

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This supplement contains Appendices A–C of the paper. Equation and table numbers with the prefix S refer to this supplement; equations, sections, and tables without the prefix refer to the main paper.

Appendix A Receiver Implementation Details

This appendix records the filter and noise-model choices needed to reproduce Section 4. Standard ECEF strapdown equations follow Groves (2013); the aircraft dynamics and control sources are cited in Section 5.

A.1 Loop Filters and Inertial Propagation

For bandwidth B_n, damping ζ, and discriminator gain k, the CT scalar DLL and PLL use

$$\omega_n = \frac{8\zeta B_n}{4\zeta^2+1}, \quad \tau_1 = \frac{k}{\omega_n^2}, \quad \tau_2 = \frac{2\zeta}{\omega_n}, \tag{S1}$$

$$u_k = u_{k-1} + \frac{\tau_2}{\tau_1}(\varepsilon_k - \varepsilon_{k-1}) + \frac{T}{\tau_1}\varepsilon_k. \tag{S2}$$

Both use $\zeta = 0.707$; DLL and PLL gains are 0.1 and 0.25, respectively. Bandwidths are listed in Table 4. The VT and UT residual PLL instead uses

$$\begin{aligned} f_{\Sigma,k} &= f_{\Sigma,k-1} + \omega_n^2 T \varepsilon_{\phi,k}, \\ f_{\mathrm{LF},k} &= f_{\Sigma,k} + 2\zeta\omega_n \varepsilon_{\phi,k}. \end{aligned} \tag{S3}$$

Its integrator is initialized to continue the CT carrier command at handover. The PLI is

$$\mathrm{PLI} = \frac{\sum_m (I_m^2 - Q_m^2)}{\sum_m (I_m^2 + Q_m^2)}, \tag{S4}$$

where I_m, Q_m are coherent 5-ms prompt sums within the 10-ms filter interval. When PLI falls below the configured threshold, both residual loop-filter states are set to zero; the navigation-derived Doppler term remains active.

The INS uses bias-corrected interval-average IMU measurements, Earth-rate compensation, finite-rotation attitude updates, and interval-average attitude for the specific-force transformation (Groves, 2013). Velocity uses gravity including J_2 and centrifugal acceleration, together with Coriolis acceleration; position uses trapezoidal integration of velocity. Between IMU epochs, position is extrapolated with the current velocity and velocity with the latest acceleration to align predictions with correlator time tags.

A.2 Error-State Dynamics and Process Noise

For the state ordering and error signs in (21), the nonzero blocks of the continuous-time matrix are

$$\begin{aligned} F_{\psi\psi} &= -\boldsymbol{\Omega}_{ie}, & F_{\psi b_g} &= \mathbf{C}_b^e, \\ F_{v\psi} &= -[\mathbf{C}_b^e \hat{\boldsymbol{f}}^b]_\times, & F_{vv} &= -2\boldsymbol{\Omega}_{ie}, \\ F_{vr} &= \boldsymbol{\Gamma}, & F_{vb_a} &= \mathbf{C}_b^e, \\ F_{rv} &= \mathbf{I}_3, & F_{b\dot{b}} &= 1. \end{aligned} \tag{S5}$$

Here $\boldsymbol{\Omega}_{ie}$ is the Earth-rate skew matrix and

$$\boldsymbol{\Gamma} = +\frac{2g_0}{r_{eS}} \frac{\boldsymbol{r}\boldsymbol{r}^\top}{\|\boldsymbol{r}\|^2}, \tag{S6}$$

with $g_0 = \|\boldsymbol{g}^e(\boldsymbol{r})\|$ and r_{eS} the ellipsoidal surface radius at the receiver latitude. This is the radial gravity-gradient term of Groves (2013, Eq. 14.49), which is written there with the downward gravitational vector, $-2\hat{\boldsymbol{\gamma}}^e\hat{\boldsymbol{r}}^\top/(r_{eS}\|\hat{\boldsymbol{r}}\|)$, and is therefore positive when expressed with g_0: a positive radial position error reduces the downward gravity and produces a positive radial velocity-error rate. At the initial position, $2g_0/r_{eS} = 3.07\times10^{-6}\ \mathrm{s}^{-2}$, in agreement with the radial derivative of the implemented gravity model obtained by central differences ($3.08\times10^{-6}\ \mathrm{s}^{-2}$). The error-state model retains this radial contribution and omits the tangential and other corrections present in the full spatial derivative of the nominal gravity model. All other blocks are zero. The implemented filter treats bias and ISB states as random walks; the simulated sensor bias instability is Gauss–Markov, as specified in Appendix B. Discretization and covariance propagation follow (29) and (30).

The inertial driving spectral densities are

$$S_g = N_g^2, \quad S_a = N_a^2, \quad S_{ba} = \frac{2\sigma_{ba}^2}{\tau_c}, \quad S_{bg} = \frac{2\sigma_{bg}^2}{\tau_c}, \tag{S7}$$

where N_g, N_a are the SI noise densities and σ_{ba}, σ_{bg} are the bias-instability standard deviations in Table 2. The additional diagonal acceleration-noise term is

$$[\mathbf{S}_u]_{jj} = \tau_u\left[(s_a|f_j|)^2 + (m_a\|\boldsymbol{f}^b\|)^2 + (k_a f_j^2)^2\right], \tag{S8}$$

where f_j is a body-axis specific-force component, s_a, m_a, k_a are the scale-factor, misalignment, and quadratic-error coefficients, and $\tau_u = 1$ s is the assumed correlation scale. The code inserts these diagonal terms directly into the ECEF velocity-noise block as an approximation. The clock and ISB densities are $S_\phi = 1\ \mathrm{m}^2/\mathrm{s}$, $S_f = 1\ \mathrm{m}^2/\mathrm{s}^3$, and $S_\beta = 10^{-4}\ \mathrm{m}^2/\mathrm{s}$.

A.3 Measurement Weighting and Initialization

With $\lambda_c = c/f_0$, linear C/N_0 in Hz, coherent interval T in seconds, and L code-discriminator outputs per update, the implemented UT range variance for one-chip early–late separation is

$$\sigma_{\rho,i}^2 = \frac{\lambda_c^2}{4LT(C/N_0)_i}\left(1 + \frac{2}{T(C/N_0)_i}\right) + \sigma_{\rho0}^2, \tag{S9}$$

where $\sigma_{\rho0} = 1$ m. The range-rate variance is

$$\begin{aligned}\sigma_{\dot\rho,i}^2 = \sigma_{\dot\rho0}^2\frac{10^{4.5}}{(C/N_0)_i}\left(1 + \frac{1}{2T(C/N_0)_i}\right)\\ + (0.02\ \mathrm{m/s})^2,\end{aligned} \tag{S10}$$

with $\sigma_{\dot\rho0} = 0.05$ m/s. Channel-selection fallback and innovation gating follow Section 4; the masks are not strict exclusion rules when fewer than four channels would remain.

The initial UT covariance is diagonal. Standard deviations are 0.05° for attitude, 1 m/s for velocity, 10 m for position, $300\ \mu$g for accelerometer bias, 0.05°/h for gyro bias, 10 m for clock bias, 1 m/s for clock drift, and 1 m for ISB. Bias estimates start at zero, while position, velocity, clock, and ISB estimates are inherited from CT. CT uses a 30-m initial ISB standard deviation. VT uses the UT values for velocity, position, clock, and ISB, and 1 m/s^2 for acceleration. The prescribed truth-based attitude initialization is stated in Section 5.

Appendix B IMU Measurement Model

The truth specific force is obtained from the ECEF trajectory as

$$\boldsymbol{f}^b = \mathbf{C}_e^b\left(\boldsymbol{a}^e + 2\boldsymbol{\Omega}_{ie}\boldsymbol{v}^e - \boldsymbol{g}^e\right), \tag{S11}$$

where gravity includes centrifugal acceleration. The inertial angular rate is

$$\boldsymbol{\omega}_{ib}^b = \boldsymbol{\omega}_{eb}^b + \mathbf{C}_e^b\boldsymbol{\omega}_{ie}^e, \tag{S12}$$

where $\boldsymbol{\omega}_{eb}^b$ is the body angular rate relative to ECEF, supplied by the attitude evolution in the fixed tangent frame. Both signals are evaluated on the 4-kHz synthesis grid.

For either sensor, the interval-average output is modeled as

$$\tilde{\boldsymbol{y}}_k = \mathcal{Q}\Bigg\{ \frac{1}{\tau_s}\int_{t_{k-1}}^{t_k} \mathrm{sat}_{y_{\max}}\big[(\mathbf{I}+\mathbf{M})\boldsymbol{y}+\boldsymbol{n}_y\big]dt + \boldsymbol{b}_0 + \boldsymbol{b}_{\mathrm{GM},k} + \boldsymbol{w}_k \Bigg\}, \tag{S13}$$

where $\boldsymbol{y}$ is specific force or angular rate, $\mathbf{M}$ contains scale-factor and cross-axis errors, and $\boldsymbol{b}_0$ is turn-on bias. Saturation precedes integration; Simpson's rule evaluates each 5-ms interval. Parameters are listed in Table 2. The implemented nonlinear terms differ between sensors:

$$\begin{aligned} \boldsymbol{n}_f &= \boldsymbol{k}_a \circ \boldsymbol{f}^b \circ \boldsymbol{f}^b, \\ \boldsymbol{n}_\omega &= \frac{\boldsymbol{k}_{\mathrm{NL},g}}{\omega_{\max}} \circ \boldsymbol{\omega}_{ib}^b \circ |\boldsymbol{\omega}_{ib}^b|, \end{aligned} \tag{S14}$$

where $\circ$ denotes componentwise multiplication. Sensor-error coefficients are independent zero-mean Gaussian draws with the specified one-standard-deviation scales.

White noise and bias instability follow

$$\boldsymbol{w}_k \sim \mathcal{N}\left(\mathbf{0}, \frac{N^2}{\tau_s}\mathbf{I}_3\right), \tag{S15}$$

$$\begin{aligned} \boldsymbol{b}_{\mathrm{GM},k} &= e^{-\tau_s/\tau_c}\boldsymbol{b}_{\mathrm{GM},k-1} \\ &\quad + \sigma_{\mathrm{BI}}\sqrt{1-e^{-2\tau_s/\tau_c}}\boldsymbol{\xi}_k, \quad \boldsymbol{\xi}_k \sim \mathcal{N}(\mathbf{0},\mathbf{I}_3), \end{aligned} \tag{S16}$$

with $\tau_c = 300$ s and an initial bias-instability draw from its stationary distribution. Noise densities are converted to SI units before use.

The quantizer $\mathcal{Q}$ retains the residual from the preceding sample. Gyroscope quantization is applied to angle increments with a $0.05\,\mu$rad step; accelerometer quantization uses a velocity-increment step equal to $0.1\,\mu$g multiplied by τ_s. The quantized increments are divided by τ_s to form the reported interval-average rates. This distinction avoids interpreting the gyro angle resolution as an angular-rate resolution.

Appendix C Evaluation Windows and Solution Availability

Table S1 lists, for each profile, the designated maneuver phase shaded in the figures and the statistical window used for the error statistics of the main paper. The window starts at the later of the phase start and 3.1 s (1 s after the VT/UT handover) and ends at the earlier of 1 s after the phase end and 59.7 s (the end of the shortest record), so it includes up to 1 s of recovery beyond the shaded phase. The baseline statistics use the whole common interval.

Table S1: Maneuver phases and statistical windows (s).

Profile	Maneuver phase (shaded)	Statistical window
Baseline (level flight)	whole record, 0–60	3.1–59.7
Turn	12.000–42.378	12.000–43.378
Loop	12.000–54.935	12.000–55.935
Break	12.000–42.286	12.000–43.286
Jinking	12.000–46.002	12.000–47.002
Rolls	15.000–34.983	15.000–35.983
Split-S/Immelmann	12.000–52.885	12.000–53.885
Departure	12.000–60.000	12.000–59.700 (clipped)

Table S2 summarizes the epoch audit of the saved navigation solutions. For every receiver, every configuration (GPS alone, NavIC alone, GPS + NavIC), and every profile, the number of epochs with finite position and velocity inside the statistical window equals the number expected from the output rate (1 ms for CT, 10 ms for VT and UT). No epoch is missing or non-finite, no NaN is omitted from the statistics, the largest gap equals one output interval, and every solution remains finite to the end of the record. The counts are identical in the three configurations.

Table S2: Expected and available finite position/velocity epochs in each statistical window; identical for GPS alone, NavIC alone, and GPS + NavIC.

Profile	CT (1 ms)	VT (10 ms)	UT (10 ms)
Baseline	56 601 / 56 601	5661 / 5661	5661 / 5661
Turn	31 379 / 31 379	3138 / 3138	3138 / 3138
Loop	43 936 / 43 936	4394 / 4394	4394 / 4394
Break	31 287 / 31 287	3129 / 3129	3129 / 3129
Jinking	35 003 / 35 003	3501 / 3501	3501 / 3501
Rolls	20 984 / 20 984	2099 / 2099	2099 / 2099
Split-S/Immelmann	41 886 / 41 886	4189 / 4189	4189 / 4189
Departure	47 701 / 47 701	4771 / 4771	4771 / 4771

Entries: available finite epochs / expected epochs.

Appendix References

Groves, P. D. (2013). *Principles of GNSS, inertial, and multisensor integrated navigation systems* (2nd ed.). Artech House.